

Status note: the Levitina–Sukochev–Zanin sequence-space conjecture

fa_banach_001 literature-status packet

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Status. Literature already answered. This packet does not claim a new proof. It records that the conjecture extracted from arXiv:1812.03861 is answered explicitly by a later paper.

Source question. The source paper is Leonard Cadilhac, *Majorization, Interpolation and non-commutative Khintchine inequalities*, arXiv:1812.03861. In the introduction, a conjecture labeled Levitina–Sukochev–Zanin quotes the Levitina–Sukochev–Zanin conjecture:

Let E be a quasi-Banach symmetric sequence space. Then E is right-2-monotone if and only if there exists $p \in (0, 2]$ such that E is an interpolation space between ℓ^p and ℓ^2 .

Cadilhac’s paper solves the corresponding function-space statement and records the sequence-space obstruction: the reverse implication for sequence spaces was tied to Calderon-type information for (ℓ^p, ℓ^q) .

Supporting answer. The supporting paper is Leonard Cadilhac, Fedor Sukochev, and Dmitriy Zanin, *Lorentz–Shimogaki–Arazy–Cwikel Theorem revisited*, arXiv:2009.02145. Its abstract says that, as a byproduct, it solves the conjecture of Levitina and the last two authors. The introduction further states that the authors resolve the conjecture of [3] in the affirmative and then gives the precise theorem.

The relevant theorem is Theorem 1.2 of arXiv:2009.02145. It states that if $E \subset \ell_\infty$ is a quasi-Banach sequence space and $q \geq 1$, then the following are equivalent:

1. there exists $p < q$ such that $E \in \text{Int}(\ell^p, \ell^q)$;
2. there is a constant $c > 0$ such that, for all $u \in E$ and $v \in \ell_\infty$, tail-majorization $|v|^q \prec_{\text{tl}} |u|^q$ implies $v \in E$ and $\|v\|_E \leq c\|u\|_E$;
3. the same tail-majorization implication holds without explicitly stating the norm bound.

Specialization to the extracted conjecture. Set $q = 2$ in Theorem 1.2. Cadilhac–Sukochev–Zanin’s tail-majorization condition $|v|^2 \prec_{\text{tl}} |u|^2$ is the same orientation as right-2-monotonicity in arXiv:1812.03861: the source paper writes that u^2 right-majorizes v^2 when all tails of u^2 dominate the corresponding tails of v^2 . Thus Theorem 1.2 gives exactly:

$$E \text{ is right-2-monotone} \iff \exists p < 2, E \in \text{Int}(\ell^p, \ell^2).$$

This is slightly sharper in the upper endpoint than the source conjecture’s $p \in (0, 2]$, since $p = 2$ is unnecessary in the theorem’s conclusion.

Provenance. The answer is not an original result of this run. The supporting authors explicitly knew they were resolving the Levitina–Sukochev–Zanin conjecture. The run contribution is the identification that the later Theorem 1.2 fully answers the source conjecture quoted in arXiv:1812.03861.

References

- [1] L. Cadilhac, *Majorization, Interpolation and noncommutative Khintchine inequalities*, arXiv:1812.03861.
- [2] L. Cadilhac, F. Sukochev, and D. Zanin, *Lorentz-Shimogaki-Arazy-Cwikel Theorem revisited*, arXiv:2009.02145.
- [3] G. Levitina, F. Sukochev, and D. Zanin, *Cwikel estimates revisited*.